Brainstorming & Idea Prioritization Template

Date	3 july 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	**Team Member**	**Idea / Suggestion**	**Category**	**Group No.**
1	Gollapalli Nimruja Lakshmi Siva Ganga	Build an ML-powered web application to predict whether a loan applicant is eligible for approval or rejection	ML / Web App	1
2	Gollapalli Nimruja Lakshmi Siva Ganga	Use Decision Tree and Random Forest classifiers to identify patterns in loan applicant data	Machine Learning	1
3	Gollapalli Nimruja Lakshmi Siva Ganga	Perform Exploratory Data Analysis using Matplotlib and Seaborn to visualize income, loan amount, and credit history trends	Data Visualization	2
4	Gollapalli Nimruja Lakshmi Siva Ganga	Preprocess the dataset by handling missing values		2

		using mean/mode imputation and encoding categorical features	Data Preprocessing	
5	Gollapalli Nimruja Lakshmi Siva Ganga	Train and compare KNN and XGBoost models; select XGBoost as the best-performing model and save it as model.pkl	Model Evaluation	3
6	Gollapalli Nimruja Lakshmi Siva Ganga	Deploy the trained XGBoost model using Flask with a responsive HTML/CSS form and result display page	Deployment / Flask	3

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	ML Model Development – Train Decision Tree, Random Forest, KNN, and XGBoost on the loan dataset; compare all four models and select the best one for deployment	High	High	1	Yes

2	Data Analysis & Preprocessing – Conduct EDA using Matplotlib/Seaborn, handle missing values with mean/mode, and encode categorical variables for model-ready input	High	High	2	Yes
3	Flask Web Deployment – Integrate the saved XGBoost model (model.pkl) into a Flask application with a user-friendly UI for real-time loan eligibility prediction	High	High	3	Yes

How the groups map to the project:

- **Group 1** — Core ML work (the intelligence of the system)
- **Group 2** — Data foundation (clean data = accurate predictions)
- **Group 3** — Flask deployment (the visible product the end-user interacts with)

All three are **Selected: Yes** because they are interdependent pillars of the Smart Lender system.